ISyE 7406 – Fall 2026

Project Proposal

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Team Project: Number 105

Project Title: Predicting Remittance Transfer Costs from the United States to Latin America and the Caribbean

Problem Statement

International remittances from the United States to Latin America and the Caribbean (LAC) represent a critical financial lifeline for millions of Latino families. Despite their importance, transfer costs remain consistently high. The United Nations set a target of 3% average cost per transaction under SDG 10.c[1], yet the corridors in this study average 5.44% as of 2025 Q1, with some destinations such as Cuba (12.0%) and Guyana (8.7%) far exceeding that benchmark.

The goal of this project is to identify which transfer and corridor characteristics best predict the total cost of a US-to-LAC remittance transaction, and to determine whether cost burden is systematically associated with specific provider types, destinations, or payment instruments. Using transaction-level data from the World Bank Remittance Prices Worldwide (RPW) database[2], we will build and compare predictive models for transfer cost and surface which corridors and provider configurations are consistently the most expensive.

Proposed Dataset and Data Preprocessing

The dataset contains approximately 198,000 transaction-level records across global remittance corridors from 2011 Q1 to 2025 Q1. Each row represents one provider offering on one corridor in one quarter, and includes: source and destination country, provider name, provider type (Bank, Money Transfer Operator, etc.), payment instrument, access point, transfer speed, flat fee in local currency, exchange rate margin, and total cost as a percentage of the amount sent.

We will subset the data to US-to-LAC corridors, yielding observations across 16 destination countries spanning 2011–2025. The target variable is `cc1 total cost %`, which captures the cost of sending as a share of the transfer amount.

[1] https://w3.unece.org/SDG/en/Indicator?id=126
[2] https://remittanceprices.worldbank.org/data-download

Key preprocessing steps:

- **Schema harmonization:** The dataset is split across two sheets with a schema change at Q2 2016. The pre-2016 sheet is missing payment instrument, access point, and receiving network coverage. We will restrict regression and classification modeling to post-Q2 2016 observations and use the full dataset for descriptive analysis only.
- **Transparency filter:** Some rows are flagged as non-transparent (exchange rate not disclosed). We will exclude these from modeling.
- **Provider type consolidation:** Credit unions (5 rows) and Non-Bank FIs (28 rows) are too sparse to model. We will retain Bank and Money Transfer Operator as the two primary categories.
- **Payment instrument standardization:** Multi-value entries (e.g., "Bank account transfer, Cash") will be simplified to their primary instrument.
- **Temporal feature engineering:** We will extract year and quarter from the period field to capture cost trends over time.
- **Encoding:** Categorical variables will be one-hot encoded for regression and classification models.

Proposed Methodology

We treat the core problem as a regression task, with cc1 total cost % as the continuous target. We will also run a secondary classification task by binarizing the target at the 3% SDG threshold (compliant vs. non-compliant). A third unsupervised task clusters corridors by their cost profile.

Exploratory Data Analysis: Before modeling, we will plot the distribution of total cost by provider type, destination country, payment instrument, and year. This will reveal skewness, outliers, and which variables show the clearest marginal relationships with cost.

1. Regression (primary task): We compare 4 models on the same feature set:
    - Multiple Linear Regression interpretable baseline.
    - Ridge Regression handles multicollinearity from one-hot encoded country and instrument variables.
    - LASSO, performs implicit feature selection, useful given the number of categorical dummies.
    - Random Forest Regression captures nonlinear interactions (e.g., provider type may matter more in some corridors than others).

2. Classification (secondary task): We binarize the target at 3% and apply logistic regression, LDA, and Random Forest classification. Beyond accuracy, we report precision, recall, and the confusion matrix broken out by provider type and destination.

Clustering (descriptive task): We aggregate the 16 destination countries by mean total cost, mean FX margin, mean fee, and transparency rate, then apply k-means and hierarchical clustering to identify which corridors share a cost profile.

Train/test split: We use a temporal split, pre-2023 for training, 2023 onward for testing, to simulate a realistic out-of-sample setting and avoid data leakage.

Planned Evaluation Strategies

For the regression models, we will use the following metrics:

- RMSE (Root Mean Squared Error): Primary metric. Penalizes large prediction errors more heavily, which matters here since outlier corridors like Cuba have costs nearly three times the mean.
- MAE (Mean Absolute Error): Secondary metric. Easier to interpret, the average error in percentage points.
- R^2 (Coefficient of Determination): Indicates how much of the variance in transfer cost is explained by the model features.

For classification, we will report accuracy, precision, recall, and F1-score, with the confusion matrix broken out by provider type and destination to check whether the model systematically misclassifies certain corridors.

For all models, we will plot residuals against key predictors (destination, provider type, year) to identify systematic under- or over-prediction. Feature importance from the Random Forest models will be used to interpret which variables drive cost most, and LASSO coefficients will provide a complementary view of variable selection.